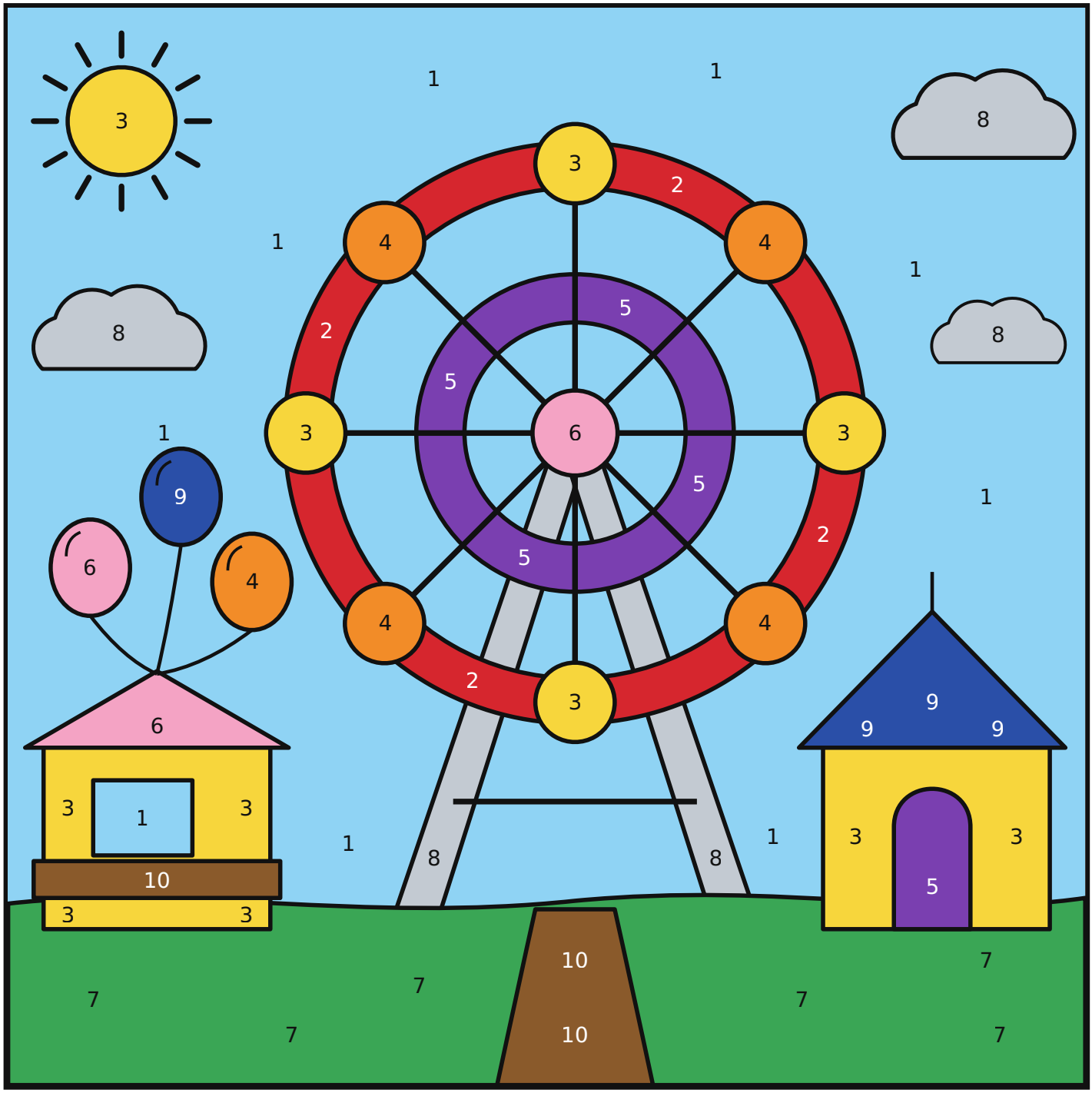
Name: **ANSWER KEY** _____ Date: _____

Circles: Color By Number



Color Code:

5 Purple 70°	**7** Green 2π	**3** Yellow 49π	**2** Red 36π	**10** Brown 9
8 Gray 6π	**9** Blue 8	**4** Orange 40°	**1** Light Blue 10π	**6** Pink 12

Name: **ANSWER KEY** _____ Date: _____

Circles: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
Find the circumference of a circle with radius 5. Leave your answer in terms of π.

$C = 2\pi r = 2\pi(5)$
10π

1. Color: **Light Blue**

2.
Find the area of a circle with radius 6. Leave your answer in terms of π.

$A = \pi r^2 = \pi(6^2)$
36π

2. Color: **Red**

3.
Find the area of a circle with diameter 14. Leave your answer in terms of π.

$r = 7$
$A = \pi(7^2)$
49π

3. Color: **Yellow**

4.
A central angle measures 80°. Find the measure of an inscribed angle that intercepts the same arc.

$\text{inscribed angle} = \frac{1}{2} \cdot \text{arc}$
40°

4. Color: **Orange**

5.
An inscribed angle measures 35°. Find the measure of its intercepted arc.

$\text{arc} = 2 \cdot \text{inscribed angle}$
70°

5. Color: **Purple**

6.
Point P is 13 units from the center O of a circle with radius 5. A tangent from P touches the circle at T. Find PT.

the radius OT is perpendicular to the tangent PT
$PT^2 + 5^2 = 13^2$
12

6. Color: **Pink**

7.
Find the length of an arc with radius 9 and central angle 40°. Leave your answer in terms of π.

$\text{arc} = \frac{40}{360} \cdot 2\pi(9)$
2π

7. Color: **Green**

8.
Find the area of a sector with radius 6 and central angle 60°. Leave your answer in terms of π.

$\text{sector} = \frac{60}{360} \cdot \pi(6^2)$
6π

8. Color: **Gray**

9.
Two chords intersect inside a circle. One chord is split into pieces of length 3 and x, and the other into pieces of length 4 and 6. Find x.

$3 \cdot x = 4 \cdot 6$
$3x = 24$
8

9. Color: **Blue**

10.
A circle has circumference 18π. Find its radius.

$2\pi r = 18\pi$
9

10. Color: **Brown**